LECTURE SERIES · DATABASE SYSTEMS

ER Modeling & Schema Design

From Real-World Concepts to Relational Tables

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ER Model · ER Design · Reduction to Schemas · Extended ER Features

MODELING THE REAL WORLD

The Entity–Relationship Model

An ER model is a high-level, visual blueprint of data — it captures the things we store, their properties, and how they connect, before any tables exist.

Entity

A real-world object or concept we store data about.

`Student, Course, Book`

Attribute

A property that describes an entity.

`Name, Age, Price`

Relationship

An association between two or more entities.

`Student Enrolls Course`

ER DIAGRAM NOTATION



rectangle *oval* *diamond* *double box* *underlined*

THE PROPERTIES OF AN ENTITY

Attributes

Attributes describe entities. They come in several flavours — knowing the type tells you how it maps to a table later.

Simple vs Composite

Atomic, or divisible into smaller parts.

```
Name → First + Last
```

Single- vs Multivalued

Holds one value, or a set of values.

```
PhoneNumbers { … }
```

Derived

Computed from other stored data.

```
Age  ←  DateOfBirth
```

Key Attribute

Uniquely identifies an entity (underlined).

```
StudentID
```

HOW ENTITIES CONNECT

Relationships & Cardinality

A relationship links entities. Cardinality says how many can take part; participation says whether every entity must.

CARDINALITY RATIOS

1 : 1 **One-to-One** One entity relates to exactly one other.

`a Person ↔ a Passport`

1 : N **One-to-Many** One relates to many on the other side.

`a Dept → many Students`

M : N **Many-to-Many** Many on both sides relate freely.

`Students ↔ Courses`

PARTICIPATION

Total (double line) — every entity must take part.

Partial (single line) — participation is optional.

DEGREE

Unary recursive — one entity	`Employee manages Employee`
Binary two entities (most common)	`Student Enrolls Course`
Ternary three entities at once	`Supplier–Part–Project`

BUILDING THE DIAGRAM

ER Model Design

DESIGN STEPS

1. **Identify entities**

   the key objects to store data about

2. **Add attributes & keys**

   the properties describing each entity

3. **Define relationships**

   how the entities associate with each other

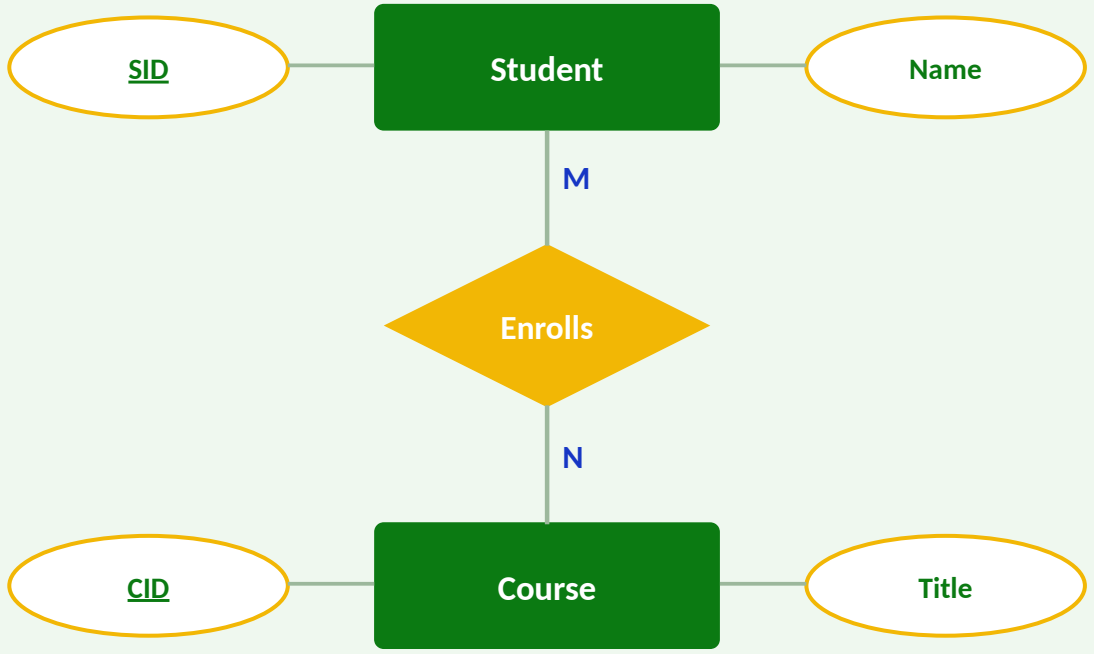
4. **Set cardinality & participation**

   how many, and whether required

5. **Draw & refine**

   assemble, review and simplify the diagram

Sample ER Diagram



ER DIAGRAM → RELATIONAL TABLES

Reduction to Schemas

Every ER diagram converts to relational tables through a small set of mechanical mapping rules.

Strong entity

Becomes its own table; its key is the primary key.

Weak entity

A table that includes the owner's key inside its primary key.

1:1 / 1:N

No new table — post a foreign key on one side.

M:N

A new table holding both keys as a composite primary key.

Multivalued attribute

Moves to a separate table linked by the entity's key.

Composite attribute

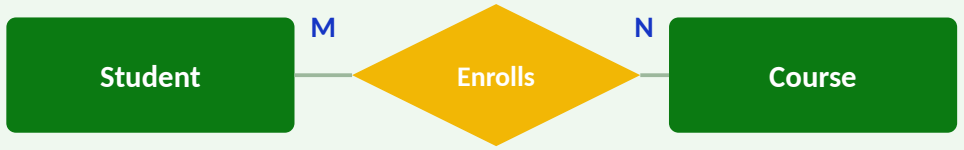
Becomes one column per component part.

MAPPING AN M:N RELATIONSHIP

Reduction in Action

Take the Student–Enrolls–Course diagram from earlier and apply the rules — the M:N relationship becomes its own table.

ER FRAGMENT



Enrolls carries an attribute: grade

a Many-to-Many relationship

RELATIONAL SCHEMA

Student

`SID (PK), Name`

Course

`CID (PK), Title`

Enrolls

`SID* , CID* (PK), grade`

M:N → new junction table with a composite key.

MODELING RICHER STRUCTURES

Extended ER (EER) Features

The Enhanced ER model adds object-oriented ideas — hierarchies and higher-level abstractions — for complex domains.

Specialization

Top-down: split an entity into sub-classes.

```
Employee → Engineer, Manager
```

Generalization

Bottom-up: merge entities into a super-class.

```
Car, Truck → Vehicle
```

Inheritance (ISA)

Sub-classes inherit the super-class attributes.

```
Engineer IS-A Employee
```

Aggregation

Treat a whole relationship as one higher-level entity.

```
(Works-On) → Manages
```

KEY TAKEAWAYS

Recap

ER Model

Entities, attributes & relationships

Attributes

Simple, composite, multivalued, derived, key

Relationships

Cardinality (1:1 / 1:N / M:N) & participation

ER Design

Identify, relate, constrain, draw

Reduction to Schemas

ER diagrams → relational tables

Extended ER

Specialization, generalization, aggregation

Thank You

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